Average Causal Effect in Observational Studies I

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Super population framework for observational studies

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- Causal quantities (estimands)
  - ACE: $\tau = \mathbb{E}\{Y_i(1) - Y_i(0)\}$
  - Average causal effect on the treated (ATT): $\tau_{\mathsf{T}} = \mathbb{E}\{Y_i(1) - Y_i(0) \mid Z_i = 1\}$
  - Average causal effect on the untreated (ATU): $\tau_{\mathsf{C}} = \mathbb{E}\{Y_i(1) - Y_i(0) \mid Z_i = 0\}$

Difficulties of observational studies

- Observational studies $\rightsquigarrow$ No randomization of treatment assignment
  - treatment assignment mechanism is often unknown
  - selection bias: $\mathbb{E}(Y_i(z) \mid Z_i = 1) - \mathbb{E}(Y_i(z) \mid Z_i = 0)$

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What is your identification assumption/strategy?

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What is your identification assumption/strategy?

  - definition of target quantity precedes identification
  - identification precedes inference

Identification

- A parameter is identifiable if it can be written as a function of the distribution of the observed data under certain model assumptions

- A parameter is nonparametrically identifiable if it can be written as a function of the distribution of the observed data without any parametric model assumptions

Nonparametric identification vs. parametric identification

- A general principle of applied statistics (Cox and Donnelly)

If an issue can be addressed nonparametrically then it will often be better to tackle it parametrically; however, if it cannot be resolved nonparametrically then it is usually dangerous to resolve it parametrically. (p. 96)

Sufficient conditions for identification

- Selection bias is zero conditional on observed covariates $\mathbf{X}_i$

$$\begin{aligned}
\mathbb{E}\{Y_i(0) \mid Z_i = 1, \mathbf{X}_i\} &= \mathbb{E}\{Y_i(0) \mid Z_i = 0, \mathbf{X}_i\} \\
\mathbb{E}\{Y_i(1) \mid Z_i = 1, \mathbf{X}_i\} &= \mathbb{E}\{Y_i(1) \mid Z_i = 0, \mathbf{X}_i\}
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- ACE is nonparametrically identifiable

$$\begin{aligned} \tau &= \mathbb{E}\{\mathbb{E}(Y_i(1) - Y_i(0) \mid \mathbf{X}_i)\} \\ &= \mathbb{E}\{\mathbb{E}(Y_i \mid Z_i = 1, \mathbf{X}_i) - \mathbb{E}(Y_i \mid Z_i = 0, \mathbf{X}_i)\} \\ &= \int \{\mathbb{E}(Y_i \mid Z_i = 1, \mathbf{X}_i = x) - \mathbb{E}(Y_i \mid Z_i = 0, \mathbf{X}_i = x)\} f(x) \mathrm{d}x \end{aligned}$$

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  - first established by Rosenbaum and Rubin (1983), also called g-formula by Robins

Ignorability

- Ignorability: $Z_i \perp\!\!\!\perp Y_i(z) \mid \mathbf{X}_i$
  - given $\mathbf{X}_i$, the observational study is like a randomized experiment
  - also known as selection on observables, unconfoundedness, conditional independence

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- Strong ignorability: $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid \mathbf{X}_i$
  - useful for causal effects on other scales
  - difference between ignorability and strong ignorability is rather technical

Plausibility of ignorability

- Example 1:

$$Y_i(1) = f_1(X_i, \epsilon_{i1}), \quad Y_i(0) = f_0(X_i, \epsilon_{i0}), \quad Z_i = \mathbf{1}\{g(X_i, \epsilon_i) \geq 0\}$$

- Example 2:

$$\begin{aligned} Y_i(1) &= f_1(X_i, U_i, \epsilon_{i1}), \quad Y_i(0) = f_0(X_i, U_i, \epsilon_{i0}) \\ Z_i &= \mathbf{1}\{g(X_i, U_i, \epsilon_i) \geq 0\} \end{aligned}$$

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- Ignorability holds in Example 1 but not 2
- Confounders: common causes of treatment and outcome
    - Ignorability means $\mathbf{X}_i$ is enough to control the treatment-outcome confounding
    - ignorability can be reasonable if we observe a rich set of covariates

Estimation: stratification

- $\mathbf{X}_i \in \{1, \ldots, K\}$
- Ignorability $Y_i(z) \perp\!\!\!\perp Z_i \mid \mathbf{X}_i \rightsquigarrow Z_i$ is "randomized" in subgroup $\mathbf{X}_i = k$
- Average the ACE within each subgroup

$$\widehat{\tau} = \sum_{k=1}^{K} \frac{n_{[k]}}{n} \cdot \widehat{\tau}_{[k]}$$

- Difficulties of the method
  - for large K, it is likely the treatment or control group in some subgroup is empty and $\widehat{\tau}_{[k]}$ is not well defined
  - it is hard to apply this stratification method to continuous $\mathbf{X}_i$

Estimation: outcome regression

- Linear outcome models: $Y_i = \beta_0 + \beta_Z Z_i + \mathbf{X}_i^\top \beta_X + \epsilon_i$

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- Run $\mathrm{lm}(Y \sim Z + X)$ and apply the ACE formula

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- $\tau(\mathbf{X}_i) = \beta_Z$ and $\tau = \beta_Z$
- Run $\text{lm}(Y \sim Z + X)$ and apply the ACE formula
- What variance estimator should be used?

Estimation: outcome regression

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- Linear outcome models with interaction:
  $Y_i = \beta_0 + \beta_Z Z_i + \mathbf{X}_i^\top \beta_X + Z_i \mathbf{X}_i^\top \beta_{ZX} + \epsilon_i$
- $\tau(\mathbf{X}_i) = \beta_Z + \beta_{ZX}^\top \mathbf{X}_i$ and $\tau = \beta_Z + \beta_{ZX}^\top \mathbb{E}(\mathbf{X}_i)$
- Run lm($Y \sim Z + X + Z * X$) and apply the ACE formula

Linear models for potential outcomes

- Linear outcome models

$$\begin{cases} Y_i(1) = \alpha_1 + \boldsymbol{X}_i^\top \gamma_1 + \epsilon_{i1}, & \mathbb{E}(\epsilon_{i1} \mid \mathbf{X}_i) = 0 \\ Y_i(0) = \alpha_0 + \boldsymbol{X}_i^\top \gamma_0 + \epsilon_{i0}, & \mathbb{E}(\epsilon_{i0} \mid \mathbf{X}_i) = 0 \end{cases}$$

$$\Longrightarrow \quad \begin{cases} \mathbb{E}\{Y_i(1) \mid \mathbf{X}_i\} = \mathbb{E}(Y_i \mid Z_i = 1, \mathbf{X}_i) = \alpha_1 + \boldsymbol{X}_i^\top \gamma_1 \\ \mathbb{E}\{Y_i(0) \mid \mathbf{X}_i\} = \mathbb{E}(Y_i \mid Z_i = 0, \mathbf{X}_i) = \alpha_0 + \boldsymbol{X}_i^\top \gamma_0 \end{cases}$$

- $\tau = \mathbb{E}[\mathbb{E}\{Y_i(1) - Y_i(0) \mid \mathbf{X}_i\}] = (\alpha_1 - \alpha_0) + (\gamma_1 - \gamma_0)^\top \mathbb{E}(\mathbf{X}_i)$

- Observed outcome ($\epsilon_i = Z_i \epsilon_{i1} + (1 - Z_i)\epsilon_{i0}$)

$$Y_i \quad = \quad \alpha_0 + (\alpha_1 - \alpha_0) \cdot Z_i + \boldsymbol{X}_i^\top \gamma_0 + Z_i \boldsymbol{X}_i^\top (\gamma_1 - \gamma_0) + \epsilon_i$$

- Run $\text{lm}(Y \sim Z + X + Z \times X)$ and apply the ACE formula with robust variance

Logistic regression for binary outcome

$$\begin{cases} Y_i(1) \mid \mathbf{X}_i \overset{ind}{\sim} \text{Bernoulli}\left(\frac{e^{\alpha_1 + \mathbf{X}_i^\top \gamma_1}}{1+e^{\alpha_1 + \mathbf{X}_i^\top \gamma_1}} \right) \\ Y_i(0) \mid \mathbf{X}_i \overset{ind}{\sim} \text{Bernoulli}\left(\frac{e^{\alpha_0 + \mathbf{X}_i^\top \gamma_0}}{1+e^{\alpha_0 + \mathbf{X}_i^\top \gamma_0}} \right) \end{cases}$$

$$\Longrightarrow \begin{cases} \text{pr}\{ Y_i(1) = 1 \mid \mathbf{X}_i \} = \text{pr}(Y_i = 1 \mid Z_i = 1, \mathbf{X}_i) = \frac{e^{\alpha_1 + \mathbf{X}_i^\top \gamma_1}}{1+e^{\alpha_1 + \mathbf{X}_i^\top \gamma_1}} \\ \text{pr}\{ Y_i(0) = 1 \mid \mathbf{X}_i \} = \text{pr}(Y_i = 1 \mid Z_i = 0, \mathbf{X}_i) = \frac{e^{\alpha_0 + \mathbf{X}_i^\top \gamma_0}}{1+e^{\alpha_0 + \mathbf{X}_i^\top \gamma_0}} \end{cases}$$

- $\tau = \mathbb{E}\left(\frac{e^{\alpha_1 + \mathbf{X}_i^\top \gamma_1}}{1+e^{\alpha_1 + \mathbf{X}_i^\top \gamma_1}} - \frac{e^{\alpha_0 + \mathbf{X}_i^\top \gamma_0}}{1+e^{\alpha_0 + \mathbf{X}_i^\top \gamma_0}} \right)$
- Estimator: $\widehat{\tau} = \frac{1}{n} \sum_{i=1}^n \left(\frac{e^{\widehat{\alpha}_1 + \mathbf{X}_i^\top \widehat{\gamma}_1}}{1+e^{\widehat{\alpha}_1 + \mathbf{X}_i^\top \widehat{\gamma}_1}} - \frac{e^{\widehat{\alpha}_0 + \mathbf{X}_i^\top \widehat{\gamma}_0}}{1+e^{\widehat{\alpha}_0 + \mathbf{X}_i^\top \widehat{\gamma}_0}} \right)$

Asymptotic variance calculation

- Delta method
- Bootstrap:
  1. independently sample n observations with replacement
  2. fit regression models and compute $\widehat{\tau}$
  3. repeat
- Quasi-Bayesian Monte Carlo (Zelig; King et al. 2000. *Amer. J. Political Sci*):
  1. sample (γ_1, γ_0) from $\mathcal{N}((\widehat{\gamma}_1, \widehat{\gamma}_0), \widehat{\text{var}}((\widehat{\gamma}_1, \widehat{\gamma}_0)))$
  2. compute $\widehat{\tau}$
  3. repeat

General outcome regression estimator

- We can use any outcome models to estimate the causal effects
  - build two predictors $\hat{\mu}_1(\mathbf{X}_i)$ and $\hat{\mu}_0(\mathbf{X}_i)$
  - estimator: $n^{-1}\sum_{i=1}^{n}\{\hat{\mu}_1(\mathbf{X}_i)-\hat{\mu}_0(\mathbf{X}_i)\}$

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- Machine learning tools for estimating the causal effects
  - e.g., Bayesian Additive Regression Trees (Hill, 2011)
  - intersection of machine learning and causal inference is an active research area

Discussion: outcome regression

- Results are sensitive to the specification of outcome models

- Complications and difficulties
  - computation of the point and variances estimators are not straightforward
  - *p*-hacking or model searching: researchers may just report the model with best performance

Evaluating evaluation methods (LaLonde. 1986. *Am. Econ. Rev.*)

- An influential study showing that statistical adjustments may not be sufficient for controlling selection bias
- Job training program: National Supported Work Demonstration
- Target population: ex-criminals, high-school drop-outs, etc.

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  - Estimates with PSID data:
    1. Difference-in-means: $-\$15205$ (s.e. $= 657$)
    2. Linear regression: \$752 (s.e. $= 782$)

Estimation: propensity score

$$\text{pr}\{Z_i, Y_i(1), Y_i(0) \mid \mathbf{X}_i\} = \text{pr}\{Y_i(1), Y_i(0) \mid \mathbf{X}_i\}\text{pr}\{Z_i \mid Y_i(1), Y_i(0), \mathbf{X}_i\}$$

- Outcome regression method is based on $\text{pr}\{Y_i(1), Y_i(0) \mid \mathbf{X}_i\}$
  $\rightsquigarrow$ sensitive to the choices of the outcome models

- Treatment assignment mechanism: $\text{pr}\{Z_i = 1 \mid Y_i(1), Y_i(0), \mathbf{X}_i\}$
  - strong ignorability: $\text{pr}\{Z_i = 1 \mid Y_i(1), Y_i(0), \mathbf{X}_i\} = \text{pr}(Z_i = 1 \mid \mathbf{X}_i)$
  - propensity score $e(\mathbf{X}_i) = \text{pr}(Z_i = 1 \mid \mathbf{X}_i)$

Estimation: propensity score

Theorem (propensity score as a dimension reduction tool)

Assume $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid \mathbf{X}_i$. *Then* $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid e(\mathbf{X}_i)$.

Proof: show $\text{pr}\{Z_i = 1 \mid Y_i(1), Y_i(0), e(\mathbf{X}_i)\} = \text{pr}\{Z_i = 1 \mid e(\mathbf{X}_i)\}$

Estimation: propensity score stratification

- Simple case: $e(\mathbf{X}_i) \in \{1, \ldots, K\}$
  - given $e(\mathbf{X}_i) = k$, there are enough treated and control units
  - $Z_i \perp\!\!\!\perp Y_i(z) \mid e(\mathbf{X}_i) = k$ for $k = 1, \ldots, K$

- Difficulties
  - $\mathbf{X}_i$ can be discrete, continuous, multi-dimensional
  - $e(\mathbf{X}_i)$ is unknown and needs to be estimated $\rightsquigarrow$ the estimated propensity score $\widehat{e}(\mathbf{X}_i)$ may not be discrete

Estimation: propensity score stratification

- General case
  - step 1: estimate $e(\mathbf{X}_i)$ (e.g., logistic model of Z_i on $\mathbf{X}_i$)
  - step 2: discretize $\widehat{e}(\mathbf{X}_i)$: $0 = q_0 < q_1 < \cdots < q_{K-1} < q_K = 1$, where q's are the cut-off points, e.g., K-th quantiles of $\widehat{e}(\mathbf{X}_i)$

$$\tilde{e}(\mathbf{X}_i) = \begin{cases} 1 & \text{if } 0 = q_0 < \widehat{e}(\mathbf{X}_i) \leq q_1 \\ \vdots & \vdots \\ K & \text{if } q_{K-1} < \widehat{e}(\mathbf{X}_i) \leq q_K = 1 \end{cases}$$

  - approximately, $Z_i \perp\!\!\!\perp Y_i(z) \mid \tilde{e}(\mathbf{X}_i)$, with discrete $\tilde{e}(\mathbf{X}_i)$
  - analysis identical to a stratified experiment

Discussion: propensity score stratification

- Depends on the propensity score model misspecification
  - only requires the correct ordering of the estimated propensity scores
  - more robust compared to other methods

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- Difficulties
    - choice of K is not obvious in finite sample: small K may $\rightsquigarrow$ large bias; large K $\rightsquigarrow$ no treated or control units within subgroup
    - Rosenbaum and Rubin suggest $K = 5$
    - if we analyze as a stratified experiment, we effectively ignore the fact that $\widehat{e}(\mathbf{X}_i)$ is unknown and estimated

Propensity score weighting

Theorem

Assume (1) $0 < e(\mathbf{X}_i) < 1$ *for all* $\mathbf{X}_i$*; (2)* $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid \mathbf{X}_i$.
Then

$$\begin{aligned} \mathbb{E}\{Y_i(1)\} &= \mathbb{E}\left\{\frac{Z_i Y_i}{e(\mathbf{X}_i)}\right\}, \quad \mathbb{E}\{Y_i(0)\} = \mathbb{E}\left\{\frac{(1-Z_i)Y_i}{1-e(\mathbf{X}_i)}\right\}, \\ \tau &= \mathbb{E}\left\{\frac{Z_i Y_i}{e(\mathbf{X}_i)} - \frac{(1-Z_i)Y_i}{1-e(\mathbf{X}_i)}\right\} \end{aligned}$$

Overlap condition

- A technical condition for identification
  - weights become infinity if $e(\mathbf{X}_i) = 0$ or 1
  - also required for outcome regression: $\mathbb{E}(Y_i \mid Z_i = 1, \mathbf{X}_i)$ and $\mathbb{E}(Y_i \mid Z_i = 0, \mathbf{X}_i)$ are well defined only under overlap condition

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- Each unit has non-zero probability of receiving treatment and control
  - only $Y_i(1)$ can be observed if $e(\mathbf{X}_i) = 1$
  - difficult to hold with many covariates

Propensity score weighting

- Horvitz-Thompson inverse probability weighting

$$\widehat{\tau}^{\text{ht}} = \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i Y_i}{\widehat{e}(\mathbf{X}_i)} - \frac{1}{n}\sum_{i=1}^{n} \frac{(1-Z_i)Y_i}{1-\widehat{e}(\mathbf{X}_i)}$$

- Hajek inverse probability weighting

$$\widehat{\tau}^{\text{hajek}} = \frac{\sum_{i=1}^{n} \frac{Z_i Y_i}{\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^{n} \frac{Z_i}{\widehat{e}(\mathbf{X}_i)}} - \frac{\sum_{i=1}^{n} \frac{(1-Z_i)Y_i}{1-\widehat{e}(\mathbf{X}_i)}}{\sum_{i=1}^{n} \frac{1-Z_i}{1-\widehat{e}(\mathbf{X}_i)}}$$

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- Estimate $\hat{e}(\mathbf{X}_i)$ first, then plug in $\hat{\tau}^{\text{ipw}}$ or $\hat{\tau}^{\text{hajek}}$

Variance

- variance estimator: complicated formula in Lunceford and Davidian (2004) $\rightsquigarrow$ bootstrap
- IPW estimator as the method of moments estimator:
  1. moment condition from the propensity score model (e.g., score)

$$\sum_{i=1}^{n} \left\{ \frac{Z_i}{e_\theta(\mathbf{X}_i)} - \frac{1 - Z_i}{1 - e_\theta(\mathbf{X}_i)} \right\} \frac{\partial}{\partial \theta} e_\theta(\mathbf{X}_i) \;=\; 0$$

  2. moment conditions from the weighting estimator

$$\text{Horvitz–Thompson} : \; \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i Y_i}{e_\theta(\mathbf{X}_i)} - \mu_1 = \frac{1}{n}\sum_{i=1}^{n} \frac{(1 - Z_i) Y_i}{1 - e_\theta(\mathbf{X}_i)} - \mu_0 = 0$$

$$\text{Hajek} : \; \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i(Y_i - \mu_1)}{e_\theta(\mathbf{X}_i)} \;=\; \frac{1}{n}\sum_{i=1}^{n} \frac{(1 - Z_i)(Y_i - \mu_0)}{1 - e_\theta(\mathbf{X}_i)} \;=\; 0$$

$\rightsquigarrow$ large sample variances are available

Truncation and trimming

- The IPW estimator blows up if the estimated propensity scores are close to 0 or 1

Truncation and trimming

- The IPW estimator blows up if the estimated propensity scores are close to 0 or 1
- Practical solutions
  - truncation: change the extreme estimated propensity score to α_l if $\hat{e}(\mathbf{X}_i) < \alpha_l$ and α_u if $\hat{e}(\mathbf{X}_i) > \alpha_u$
  - propensity score trimming: drop units with $\widehat{e}(\mathbf{X}_i)$ outside $[\alpha_l, \alpha_u]$
  - typical choices of $[\alpha_l, \alpha_u]$: $[0.1, 0.9]$, $[0.05, 0.95]$

True propensity score vs. estimated propensity score

- Data: (Y_i, Z_i, X_i), $i = 1, \ldots, n$, Z_i and X_i are binary
- $n_{zx} = \#\{i : Z_i = z, X_i = x\}$

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$$\widehat{e}(x) = \begin{cases} \frac{n_{10}}{n_{00}+n_{10}} & \text{if } x = 0 \\ \frac{n_{11}}{n_{01}+n_{11}} & \text{if } x = 1 \end{cases}$$

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- Which one is more efficient?
  - $\widehat{\mu}_{\text{true}}$: difference in means estimator
  - $\widehat{\mu}_{\text{est}}$: covariate adjusted estimator, post-stratification estimator

Balancing property of propensity score

Theorem

Assume $0 \leq e(\mathbf{X}_i) \leq 1$ *for all* $\mathbf{X}_i$.
Then, $Z_i \perp\!\!\!\perp \mathbf{X}_i \mid e(\mathbf{X}_i)$ *and for any* $h(\mathbf{x})$ *with* $\mathbb{E}|h(\mathbf{X}_i)| < \infty$

$$\mathbb{E}\left\{\frac{Z_i h(\mathbf{X}_i)}{e(\mathbf{X}_i)}\right\} = \mathbb{E}\left\{\frac{(1 - Z_i)h(\mathbf{X}_i)}{1 - e(\mathbf{X}_i)}\right\}$$

The central role of propensity score

- Check covariate balance

- Reasons of imbalance: misspecified propensity score model

- Design of observational studies without having access to the outcome
  $\rightsquigarrow$ objective causal inference

- How to choose $h(\mathbf{x})$

Real data: small volume hospital vs. large volume hospital

- Target population: cardia cancer patients in Sweden
- Question: whether it is better to be treated in a large or small volume hospital
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- Covariate balance check
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- Estimators for ACE:
  - Outcome regression: 0.036 (s.e. $= 0.086$)
  - HT: 0.017 (s.e. $= 0.089$)
  - Hajek: 0.034 (s.e. $= 0.088$)

Hybrid estimator

- Stratification + Outcome regression
  - $\tilde{e}(\mathbf{X}_i) \in \{1, \ldots, K\}$
  - within $\tilde{e}(\mathbf{X}_i) = k$, covariates are not exactly balanced
  - within in $\tilde{e}(\mathbf{X}_i) = k$, we can use outcome regression method to obtain $\widehat{\tau}_{\text{adj},[k]}$
  - $\widehat{\tau}_{\text{adj}} = \sum_{k=1}^{K} \frac{n_{[k]}}{n} \widehat{\tau}_{\text{adj},[k]}$

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- Weighting + Regression $\rightsquigarrow$ doubly robust estimator
  - Inverse probability weighting: $\tau = \mathbb{E}\left\{\frac{Z_i Y_i}{e(\mathbf{X}_i)}\right\} - \mathbb{E}\left\{\frac{(1-Z_i)Y_i}{1-e(\mathbf{X}_i)}\right\}$
  - outcome regression: $\tau = \mathbb{E}\{\mu_1(\mathbf{X}_i)\} - \mathbb{E}\{\mu_0(\mathbf{X}_i)\}$, where $\mu_z(\mathbf{X}_i) = \mathbb{E}\{Y_i(z) \mid \mathbf{X}_i\} = \mathbb{E}(Y_i \mid Z_i = z, \mathbf{X}_i)$
  - any convex combination of the RHS of the two equations will give the ACE

Doubly robust estimator: decomposition

$$
\begin{aligned}
& \tau \\
= \; & \underbrace{\mathbb{E}\{Y_i(1) - \mu_1(\mathbf{X}_i)\} - \mathbb{E}\{Y_i(0) - \mu_0(\mathbf{X}_i)\}}_{\text{ACE for residuals}} + \underbrace{\mathbb{E}\{\mu_1(\mathbf{X}_i)\} - \mathbb{E}\{\mu_0(\mathbf{X}_i)\}}_{\text{Outcome regression}} \\
= \; & \underbrace{\mathbb{E}\left[\frac{Z_i\{Y_i - \mu_1(\mathbf{X}_i)\}}{e(\mathbf{X}_i)}\right] - \mathbb{E}\left[\frac{(1 - Z_i)\{Y_i - \mu_0(\mathbf{X}_i)\}}{1 - e(\mathbf{X}_i)}\right]}_{\text{use IPW formula}} \\
& + \mathbb{E}\{\mu_1(\mathbf{X}_i)\} - \mathbb{E}\{\mu_0(\mathbf{X}_i)\} \\
= \; & \underbrace{\mathbb{E}\left[\frac{Z_i\{Y_i - \mu_1(\mathbf{X}_i)\}}{e(\mathbf{X}_i)} + \mu_1(\mathbf{X}_i)\right]}_{\text{for } \mathbb{E}\{Y_i(1)\}} - \underbrace{\mathbb{E}\left[\frac{(1 - Z_i)\{Y_i - \mu_0(\mathbf{X}_i)\}}{1 - e(\mathbf{X}_i)} + \mu_0(\mathbf{X}_i)\right]}_{\text{for } \mathbb{E}\{Y_i(0)\}}
\end{aligned}
$$

Doubly robust estimator

- In practice, we need to assume models for $e(\mathbf{x})$ and $\{\mu_1(\mathbf{x}), \mu_0(\mathbf{x})\}$. We can assume parametric models
  - $e(\mathbf{x}, \beta)$, e.g., logistic model of Z_i on $\mathbf{X}_i$
  - $\mu_1(\mathbf{x}, \alpha_1)$ and $\mu_0(\mathbf{x}, \alpha_0)$, e.g., linear or logistic model of $Y_i(1)$ on $\mathbf{X}_i$ and $Y_i(0)$ on $\mathbf{X}_i$
  - Doubly robust estimator

$$
\begin{aligned}
\tilde{\mu}_{1,\mathrm{DR}} &= \mathbb{E}\left[\frac{Z_i\{Y_i - \mu_1(\mathbf{X}_i, \alpha_1)\}}{e(\mathbf{X}_i, \beta)} + \mu_1(\mathbf{X}_i, \alpha_1)\right] \\
\tilde{\mu}_{0,\mathrm{DR}} &= \mathbb{E}\left[\frac{(1 - Z_i)\{Y_i - \mu_0(\mathbf{X}_i, \alpha_0)\}}{1 - e(\mathbf{X}_i, \beta)} + \mu_0(\mathbf{X}_i, \alpha_0)\right]
\end{aligned}
$$

Doubly robust estimator

- Step 1: fit a propensity score model $\rightsquigarrow e(\mathbf{x}, \widehat{\beta})$
- Step 2: fit two outcome models $\rightsquigarrow \mu_1(\mathbf{x}, \widehat{\alpha}_1)$ and $\mu_0(\mathbf{x}, \widehat{\alpha}_0)$
- Step 3:

$$
\begin{aligned}
\widehat{\mu}_{1,\mathrm{DR}} &= \frac{1}{n}\sum_{i=1}^{n}\left[\frac{Z_i\{Y_i - \mu_1(\mathbf{X}_i, \widehat{\alpha}_1)\}}{e(\mathbf{X}_i, \widehat{\beta})} + \mu_1(\mathbf{X}_i, \widehat{\alpha}_1)\right] \\
\widehat{\mu}_{0,\mathrm{DR}} &= \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1 - Z_i)\{Y_i - \mu_0(\mathbf{X}_i, \widehat{\alpha}_0)\}}{1 - e(\mathbf{X}_i, \widehat{\beta})} + \mu_0(\mathbf{X}_i, \widehat{\alpha}_0)\right] \\
\widehat{\tau}_{\mathrm{DR}} &= \widehat{\mu}_{1,\mathrm{DR}} - \widehat{\mu}_{0,\mathrm{DR}}
\end{aligned}
$$

Doubly robustness property

Theorem

If either $\mu_1(\mathbf{x}, \alpha_1) = \mu_1(\mathbf{x})$ *or* $e(\mathbf{x}, \beta) = e(\mathbf{x})$*, then* $\tilde{\mu}_{1,DR} = \mathbb{E}\{Y_i(1)\}$*;*
if either $\mu_0(\mathbf{x}, \alpha_0) = \mu_0(\mathbf{x})$ *or* $e(\mathbf{x}, \beta) = e(\mathbf{x})$*, then* $\tilde{\mu}_{0,DR} = \mathbb{E}\{Y_i(0)\}$*;*
if either $\{\mu_1(\mathbf{x}, \alpha_1), \mu_0(\mathbf{x}, \alpha_0)\} = \{\mu_1(\mathbf{x}), \mu_0(\mathbf{x})\}$ *or* $e(\mathbf{x}, \beta) = e(\mathbf{x})$*, then* $\tau = \tilde{\mu}_{1,DR} - \tilde{\mu}_{0,DR}$

Proof: show that

$$\tilde{\mu}_{1,\mathrm{DR}} - \mathbb{E}\{Y_i(1)\} \to \mathbb{E}\left[\frac{e(\mathbf{X}_i) - e(\mathbf{X}_i, \beta)}{e(\mathbf{X}_i, \beta)} \cdot \{\mu_1(\mathbf{X}_i) - \mu_1(\mathbf{X}_i, \alpha_1)\}\right]$$

Doubly robust estimator

- $\widehat{\tau}_{\mathrm{DR}} = \frac{1}{n}\sum_{i=1}^{n}\widehat{\tau}_{i,\mathrm{DR}}$

$$\begin{aligned}\widehat{\tau}_{i,\mathrm{DR}} &= \left[\frac{Z_i\{Y_i - \mu_1(\mathbf{X}_i,\widehat{\alpha}_1)\}}{e(\mathbf{X}_i,\widehat{\beta})} + \mu_1(\mathbf{X}_i,\widehat{\alpha}_1)\right] \\ &\quad - \left[\frac{(1-Z_i)\{Y_i - \mu_0(\mathbf{X}_i,\widehat{\alpha}_0)\}}{1-e(\mathbf{X}_i,\widehat{\beta})} + \mu_0(\mathbf{X}_i,\widehat{\alpha}_0)\right]\end{aligned}$$

- Variance estimator
  - doubly robust estimator as a moment estimator
  - Lunceford and Davidian (2004) $\widehat{\mathrm{var}}\left(\widehat{\tau}_{\mathrm{DR}}\right) = \frac{1}{n^2}\sum_{i=1}^{n}\left(\widehat{\tau}_{i,\mathrm{DR}} - \widehat{\tau}_{\mathrm{DR}}\right)^2$
  - bootstrap

Comparison

- Outcome regression: $\mu_1(\mathbf{x}, \alpha_1)$ and $\mu_0(\mathbf{x}, \alpha_0)$ are correctly specified
- Propensity score method: $e(\mathbf{x}, \beta)$ is correctly specified
- Doubly robust method: either $\{\mu_1(\mathbf{x}, \alpha_1), \mu_0(\mathbf{x}, \alpha_0)\}$ or $e(\mathbf{x}, \beta)$ is correctly specified

- Efficiency comparison: semiparametric efficiency theory

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  - Doubly robust: 0.040 (s.e. = 0.089)

A Simulation Study

- Simulations are doomed to succeed?

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- Estimators to be evaluated:
  1. Outcome regression
  2. Horvitz-Thompson
  3. Hajek
  4. Doubly robust

A Simulation Study

- Simulations are doomed to succeed?
- Estimators to be evaluated:
  1. Outcome regression
  2. Horvitz-Thompson
  3. Hajek
  4. Doubly robust
- Setup:
  - 2 covariates: X_{i1} and X_{i2}, standard normal
  - Outcome model: linear model
    - correct: $Y_i(1) = 1 + 2X_{i1} + X_{i2} + \epsilon_{i1}$, $Y_i(0) = 1 + 2X_{i1} + X_{i2} + \epsilon_{i2}$
    - incorrect: $Y_i(1) = 1 + 2\exp(X_{i1}) - \exp(X_{i2}) + \epsilon_{i1}$, $Y_i(0) = 1 - 2\exp(X_{i1}) + \exp(X_{i2}) + \epsilon_{i2}$
  - Propensity score model: logistic model
    - correct: $\text{logit}\{\text{pr}(Z_i = 1 \mid X_{i1}, X_{i2})\} = X_{i1} + X_{i2}$
    - incorrect: $\text{logit}\{\text{pr}(Z_i = 1 \mid X_{i1}, X_{i2})\} = -1 + \exp(X_{i1}) - \exp(X_{i2})$

Simulation result

	OR	HT	Hajek	DR
(1) Both models correct				
Bias	−0.003	0.097	0.111	0.000
Standard error	0.176	0.361	0.334	0.181
(2) Outcome model correct				
Bias	0.003	−0.690	−0.634	0.064
Standard error	0.192	0.886	0.671	0.333
(3) Propensity score model correct				
Bias	−0.441	−0.050	−0.044	−0.128
Standard error	0.624	1.030	1.003	0.877
(4) Both models incorrect				
Bias	−0.723	−0.282	−0.514	0.986
Standard error	0.869	0.615	0.719	1.692

Summary

- The identification of the average treatment effect in observational studies typically requires:
  1. overlap
  2. ignorability

- Various methods to estimate causal effects:
  - outcome regression
  - weighting methods as a generalization of matching methods
    - be careful about extreme weights
    - balancing property of propensity score covariates
  - doubly robust estimation: combine weighting and regression
    - consistent if either the propensity score model or the outcome model is correct

Suggested readings

- Identification under ignorability and outcome regression:
  - Hernan and Robins. Chapters 13.1-13.3, 15.1
- Weighting:
  - Hernan and Robins. Chapter 12.1-12.3
  - Imbens and Rubin. Chapter 17.8
- General discussion and comparison
  - Hernan and Robins. Chapter 15
  - Lunceford and Davidian. 2004. "Stratification and weighting via the propensity score in estimation of causal treatment effects: a comparative study." *Statistics in Medicine*